from a source measure to a target measure, which has various applications in visual matching (Wang et al., 2013), long-tailed learning (Shi et al., 2024b;c), time series analysis (Zhang et al., 2020; Shi et al., 2020), multi-modal learning (Shi et al., 2024a; Wang et al., 2024), and etc. Specifically, given the cost matrix $\mathbf{C}$ and two marginals $(\mathbf{a}, \mathbf{b})$, Kantorovich's OT (Kantorovich, 1960) with entropic regularization involves solving the coupling $\mathbf{P}$ by:

$$\min_{\mathbf{P}\in U(\mathbf{a},\mathbf{b})} <\mathbf{C},\mathbf{P}> -\epsilon H(\mathbf{P}), \quad \text{where} \quad U(\mathbf{a},\mathbf{b}) = \{\mathbf{P}\in\mathbb{R}^{+}_{mn}|\mathbf{P}\mathbf{1}_n = \mathbf{a}, \mathbf{P}^\top\mathbf{1}_m = \mathbf{b}\} \tag{1}$$

Note that $H(\mathbf{P}) = - < \mathbf{P}, \log\mathbf{P} - \mathbf{1}_{m\times n} >$ represents the entropic regularization, and ϵ is the coefficient. When $\epsilon = 0$, the entropic OT degenerates to vanilla Kantorovich, and when $\epsilon > 0$, the objective in Eq. 1 is ϵ-strongly convex. Consequently, it possesses a unique solution that can be determined using Sinkhorn algorithms, as discussed in (Cuturi, 2013; Benamou et al., 2015).

Proposition 1 (Solution Form of Entropic OT (Peyre & Cuturi, 2019))**.** *The solution to Eq. 1 is unique and satisfy $\mathbf{P} = diag(\mathbf{u})\mathbf{K}diag(\mathbf{v})$, where $\mathbf{K} = e^{-\mathbf{C}/\epsilon}$ is the Gibbs kernel associated to the cost matrix $\mathbf{C}$ and $(\mathbf{u}, \mathbf{v})$ are two (unknown) scaling variables.*

With the solution form of entropic OT, Sinkhorn algorithms (Cuturi, 2013; Benamou et al., 2015) were proposed to obtain the solution of Eq. 1, where one can iteratively update $\mathbf{u} = \mathbf{a}/(\mathbf{K}\mathbf{v})$ and $\mathbf{v} = \mathbf{b}/(\mathbf{K}^\top\mathbf{v})$ to approximate the solution of the transportation problem. However, research on solving operational research problems using GPU-friendly matrix iteration algorithms seems to be focused only on transportation problems or matching problems, rather than other complex problems in operation research. In this paper, we propose to use matrix-vector iterative algorithms to solve a more generalized transportation problem, i.e. MCF. To the best of our knowledge, our algorithm is the first GPU-friendly matrix iterative method specifically designed for solving MCF.

Optimal Transport on the Graph. The concept of optimal transport on graphs can be attributed to (Feldman & McCann, 2002), who initially compute the shortest distances between source and target nodes to establish a cost matrix. This matrix is then used to calculate the 1-Wasserstein distance, transforming the problem into a linear program, specifically a min-cost flow problem. This methodology has been applied and expanded to formulate and analyze traffic congestion models. Recently, (Le et al., 2022) introduced a new variation named Sobolev transport (ST), tailored for measures supported on graphs, enabling a closed-form expression for quicker computation. Furthermore, (Le et al., 2024) extended Sobolev transport with an Orlicz structure (Orlicz, 1932). They need to first calculate shortest paths before solving the optimal transport between source and target and they can't handle capacity constraints at nodes. In this paper, we introduce optimal flow transport and its entropic regularized case, which calculate the flow in the graph via matrix iterative methods without the need for precomputing the shortest distances on the graph.

Minimum Cost Flow. Machine learning has been leveraged to handle optimization problems (Bengio et al., 2021). Minimum-cost flow (MCF) (Goldberg, 1997) is one of the optimization problems from the network flow theory (Iri, 1996), which involves finding the minimum-cost transportation of the specified amount of a commodity from a set of supply nodes to a set of demand nodes in a directed network, considering capacity constraints and linear cost defined on the arcs. The MCF problem arises in diverse applications, including transportation (Ahmady & Eftekhari Yeghaneh, 2022), logistics (Krile, 2004), and many other industrial scenes (Choi et al., 1988). Efficient algorithms have been developed to solve the MCF problem. For instance, Fulkerson's out-of-kilter algorithm (Fulkerson, 1961) takes advantage of the special structure of MCF and adjusts the arcs that do not satisfy optimality properties. The cycle canceling algorithm is proposed by Klein (Klein, 1967), which focuses on maintaining feasible flow at each iteration. Recent years, advancements in the MCF problem have been marked by several pivotal studies. However, most of the previous methods is based on CPU calculations and also make an assumption that all the data is integers (de Vos, 2023), which limits the solution of the MCF problem when dealing with high-precision data. In this paper, we solve MCF by using a GPU-friendly matrix iterative method to get the approximate solutions.

3 Methdology

3.1 Optimal Flow Transport

In this subsection, We give the definition of Optimal Flow Transport and its exact methods for solving.